

FERC Order No. 901 - Reliability Standards to Address Inverter-Based Resources

FERC ORDER NO. 901 - RELIABILITY STANDARDS TO ADDRESS INVERTER-BASED RESOURCES

Docket No. RM22-12-000

Issued October 19, 2023

SUMMARY: The Federal Energy Regulatory Commission (Commission) is directing the North American Electric Reliability Corporation (NERC), the Commission-certified Electric Reliability Organization, to develop new or modified Reliability Standards that address reliability gaps related to inverter-based resources in the following areas: data sharing; model validation; planning and operational studies; and performance requirements.

I. Introduction

Pursuant to section 215(d)(5) of the Federal Power Act (FPA), the Federal Energy Regulatory Commission (Commission) directs the North American Electric Reliability Corporation (NERC), the Commission-certified Electric Reliability Organization (ERO), to submit new or modified Reliability Standards that address specific matters pertaining to the impacts of inverter-based resources (IBR) on the reliable operation of the Bulk-Power System. As proposed in the notice of proposed rulemaking (NOPR), we direct NERC to develop new or modified Reliability Standards addressing reliability gaps pertaining to IBRs in four areas: (1) data sharing; (2) model validation; (3) planning and operational studies; and (4) performance requirements.

We take this action in light of the rapid change in the mix of generation resources connecting to the Bulk-Power System, including the addition of an "unprecedented proportion of nonsynchronous resources" projected to connect over the next decade, including many generation resources that employ inverters, rectifiers, and converters to provide energy to the Bulk-Power System. According to NERC, the rapid integration of IBRs is "the most significant driver of grid transformation" on the Bulk-Power System.

The Reliability Standards, first approved by the Commission in 2007, were developed to apply to the types of generation resources prevalent at that time - nearly exclusively synchronous generation resources - to ensure the reliable operation of the Bulk-Power System. As a result, the Reliability Standards may not account for the material technological differences between the response of synchronous generation resources and the response of IBRs to the same disturbances on the Bulk-Power System.

We therefore direct NERC, pursuant to section 215(d)(5) of the FPA and section 39.5(f) of the Commission's regulations, to develop new or modified Reliability Standards that address the following specific issues:

- **IBR Data Sharing:** The Reliability Standards must require that generator owners, transmission owners, and distribution providers share validated modeling, planning, operations, and disturbance monitoring data for all IBRs with planning coordinators, transmission planners, reliability coordinators, transmission operators, and balancing authorities so that the latter group has the necessary data to predict the behavior of

registered and unregistered IBRs individually and in the aggregate, as well as IBR-DERs in the aggregate, and their impact on the reliable operation of the Bulk-Power System.

- IBR Model Validation: The Reliability Standards must require that all IBR models are comprehensive, validated, and updated in a timely manner, so that planning coordinators, transmission planners, reliability coordinators, transmission operators, and balancing authorities can adequately predict the behavior of registered and unregistered IBRs individually and in the aggregate, as well as IBR-DERs in the aggregate, and their impacts on the reliable operation of the Bulk-Power System.

- IBR Planning and Operational Studies: The Reliability Standards must require that planning and operational studies include validated IBR models to assess the reliability impacts of registered and unregistered IBRs individually and in the aggregate, as well as IBR-DERs in the aggregate, on the reliable operation of the Bulk-Power System. The Reliability Standards must require that planning and operational studies assess the impacts of all IBRs within and across planning and operational boundaries for normal operations and contingency event conditions.

- IBR Performance Requirements: The Reliability Standards must ensure that registered IBRs will provide frequency and voltage support during frequency and voltage excursions in a manner necessary to contribute toward the overall system needs for essential reliability services. The Reliability Standards must establish clear and reliable technical limits and capabilities for registered IBRs to ensure that all registered IBRs are operated in a predictable and reliable manner during normal operations and contingency event conditions. The Reliability Standards must require that the operational aspects of registered IBRs contribute towards meeting the overall system needs for essential reliability services. The Reliability Standards must include post-disturbance ramp rates and phase lock loop synchronization requirements for registered IBRs.

NERC's standards development plan must ensure that NERC submits new or modified Reliability Standards by the following deadlines. First, by November 4, 2024, NERC must submit new or modified Reliability Standards that establish IBR performance requirements, including requirements addressing frequency and voltage ride through, post-disturbance ramp rates, phase lock loop synchronization, and other known causes of IBR tripping or momentary cessation. NERC must also submit, by November 4, 2024, new or modified Reliability Standards that require disturbance monitoring data sharing and post-event performance validation for registered IBRs. Second, by November 4, 2025, NERC must submit new or modified Reliability Standards addressing the interrelated directives concerning: (1) data sharing for registered IBRs, unregistered IBRs, and IBR-DERs in the aggregate; and (2) data and model validation for registered IBRs, unregistered IBRs, and IBR-DERs in the aggregate. Finally, by November 4, 2026, NERC must submit new or modified Reliability Standards addressing planning and operational studies for registered IBRs, unregistered IBRs, and IBR-DERs in the aggregate.

II. Background - Inverter-Based Resources

The Bulk-Power System generation fleet has traditionally been composed almost exclusively of synchronous generation resources that convert mechanical energy into electric energy through electromagnetic induction. By virtue of the kinetic energy in their large rotating components, these synchronous generation resources inherently resist changes in system frequency, providing time for other governor controls (when properly

configured) to maintain supply and load balance. Similarly, synchronous generation resources inherently provide voltage support during voltage disturbances.

In contrast, IBRs do not use electromagnetic induction from machinery that is directly synchronized to the Bulk-Power System. Instead, the majority of installed IBRs use grid-following inverters, which rely on sensed information from the grid (e.g., a voltage waveform) to produce the desired AC real and reactive power output. Due to their inverters, IBRs can track grid state parameters (e.g., voltage angle) in milliseconds and react nearly instantaneously to changing grid conditions. Some IBRs, however, are not configured or programmed to support grid voltage and frequency in the event of a system disturbance, and, as a result, will reduce power output, exhibit momentary cessation, or trip in response to variations in system voltage or frequency.

III. Performance Requirements - Post-disturbance IBR Ramp Rate Interactions and Phase Lock Loop Synchronization

The currently effective Reliability Standards do not require all generation resources that momentarily cease operation following a system disturbance to return to pre-disturbance output levels without impeded ramp rates or require that all generation resources maintain voltage phase angle synchronization with the Bulk-Power System grid voltage during a system disturbance. IBRs that lose synchronization with grid voltage (i.e., phase lock loop loss of synchronism) will momentarily cease current injection into the grid during Bulk-Power System disturbance events due to protection and control settings. Such momentary cessation occurrences exacerbate system disturbances and have a material impact on the reliable operation of the Bulk-Power System.

Phase lock loop (PLL) synchronization refers to the inverter continuing to inject grid current during system disturbances, locking the phase-locked loop to the last synchronization point, and continuing to inject current into the grid at its calculated phase until the phase-locked loop can resynchronize. Maintaining PLL synchronization is important for protecting and controlling the inverter's response and for ensuring the inverter continues to support Bulk-Power System reliability during and immediately after a disturbance event.

Reliability Standard PRC-024-3 does not require registered IBRs (or any generator) to remain connected to the Bulk-Power System and to continue to inject current and support frequency inside the "no trip zone." Therefore, IBRs could continue to act adversely in response to normally cleared faults by continuing to exhibit momentary cessation and power reduction behaviors.